# CS4433 Moderated LMS – Code Review Record

## 1. Code Review Checklist

| Review area | Review criteria | Status |
|---|---|---|
| Code structure | Classes and responsibilities are organized into models, repositories and services. | Reviewed |
| Naming conventions | Classes, functions and variables use understandable names. | Reviewed |
| Encapsulation | Data members are protected through appropriate access control. | Reviewed |
| Database operations | Repository classes handle database interactions. | Reviewed |
| Input validation | User input is validated before processing. | Reviewed |
| Authentication | Password hashing and verification are implemented. | Reviewed |
| Access control | Functionality is separated according to user roles. | Reviewed |
| Assignment submissions | Submission records and statuses are handled by the system. | Reviewed |
| Late penalties | The graduated late penalty policy is applied to late submissions. | Reviewed |
| Grading and moderation | Grading, moderation and result release follow the intended workflow. | Reviewed |
| Error handling | Relevant operations handle invalid input and failures. | Reviewed |
| Testing | Automated tests were executed during quality assurance. | Tested |

## 2. Review Findings

- The source-code review and static analysis identified areas for possible improvement, including:
  - Some functions have relatively high cyclomatic complexity.
  - Some loops could be replaced with standard library algorithms.
  - A few parameters could be passed by const reference.
  - Some GoogleTest fixture members were reported as unused by Cppcheck.
- These findings were reviewed and documented. They were not all changed because some are recommendations or false positives, and unnecessary refactoring could introduce regressions.

## 3. Review Conclusion

The project was reviewed for code organisation, naming, encapsulation, database operations, input validation, authentication, assignment processing and moderated grading.

Automated testing and static analysis were also performed. The identified findings were documented for future improvement.